

Seawater-Induced Healable Underwater Superoleophobic Antifouling Coatings

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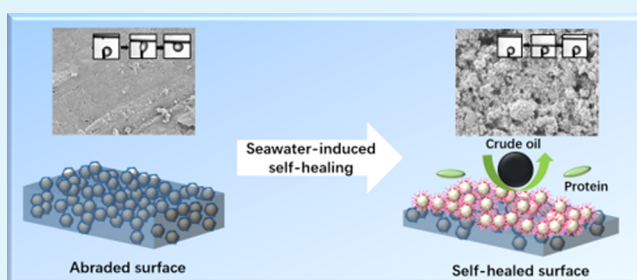
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Supporting Information

ABSTRACT: Creating an artificial surface, mimicking a live fish scale that repels oil underwater and with self-healing properties, would be significant for the development of nontoxic marine antifouling coatings. Here, we report a seawater-induced strategy to create in situ an underwater superoleophobic surface, starting from the coatings of a self-polishing polymer and seawater-responsive polymer-grafted SiO₂ nanoparticles. The coatings' surfaces were able to renew in artificial seawater through the hydrolysis of the superficial self-polishing polymer and its subsequent dissolution. Particularly, the grafted poly(triisopropylsilyl acrylate-co-3-methacryloxypropyltrimethoxysilane) chains could transform into hydrophilic ones via seawater-induced hydrolysis, which additionally strengthened the oil-repellency (zero oil adhesive force) and endowed the surface with excellent antiprotein adsorption characteristics. Because the hydrolysis was limited to the superficial layer of the coatings, it could avoid the water-swelling that instead occurs with conventional underwater superoleophobic coatings, with significant benefits to its durability. We believe that the seawater-induced renewal of underwater superoleophobic surfaces will be useful in extreme marine environments.

KEYWORDS: underwater superoleophobic coatings, antifouling, renewable, silica nanoparticles, seawater responsive polymer



1. INTRODUCTION

Marine biofouling has been a troubling phenomenon for thousands of years because it decreases ship speed and maneuverability, accelerates corrosion, increases fuel consumption, and limits the performance of many marine applications. Many strategies^{1–5} have been proposed to inhibit the biofouling of surfaces through an increased resistance toward the nonspecific adsorption of proteins, microbes, plants, and animals. The most effective and convenient method is to use antifouling coatings.^{6–8} Today, oil spills taking place during the extraction and transportation of crude oil make the marine environment more complex. The oil pollutes the antifouling coating surfaces, weakening their mechanical performance and shortening the life. Marine antifouling coatings are thus facing new challenges.^{9–12}

To resist oil pollution, significant attention has been paid to natural antifouling surfaces.^{12–15} Aquatic organisms, such as carps,¹³ whales,¹⁶ sharks,¹⁷ filefish,¹⁸ and seaweed,¹⁹ utilize reliable strategies to contrast the growth of fouling organisms and keep their skin clean under oil-polluted water. By imitating the chemical–physical attributes of fish scales, Jiang's group successfully constructed underwater superoleophobic surfaces based on the combination of a hierarchical micro/nanostructure surface with high surface energy components.¹³ Since

then, many metal oxides²⁰ and polymer hydrogels^{12,19,21,22} have been employed to fabricate underwater superoleophobic surface. A water layer is trapped on these surfaces, resulting in oil-repellency and reduced contact between the organism and the materials' surfaces; however, metal oxides are prone to corrosion in seawater, whereas hydrophilic polymers are unstable in water. Hence, fabrication techniques for superoleophobic antifouling coatings characterized by a long lifetime in the chemically complex and hydrodynamic marine environments are still required.¹⁴ Inspired by columnar nacre, Guo et al.¹⁰ adopted a template replica process to fabricate an artificial layered montmorillonite/hydroxyethyl cellulose nacre-like material, with a convex hexagonal columnar structure on its surface. This material showed superior resistance to sand grain impingement, stable underwater superoleophobicity, and low adhesion even after having been submerged in seawater for 15 days. Manna et al. developed covalent polymeric multilayer coatings to create underwater superhydrophobic surfaces.^{23–25} These coatings could tolerate a broad range of physical, chemical, and environmental challenges. Nonetheless, many

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underwater superoleophobic surfaces still present issues such as failure under mechanical forces, inability to self-heal, failure of chemical groups and efficiency loss in high-salinity environments, complex fabrication techniques, and high cost.

Live fish scales possess durable antifouling performance because of the regeneration of their micro/nanostructure and mucus. After their death, the surface of the scales gradually loses its antifouling performance because of the biodegradation of the mucus. Therefore, the best way to maintain a durable antifouling performance of the underwater superoleophobic coating is to endow it with a renewal ability similar to that of live organisms. To date, most of the proposed underwater superoleophobic materials do not have such an ability.

Recently, our group fabricated self-healing underwater superoleophobic coatings through the assembly of hierarchical microgel spheres. After having been mechanically damaged, the coatings could re-establish their surface topography and composition through the swelling of the microgel spheres and simultaneous rearrangement of the hydrophilic polymers on the surface.⁹ Here, inspired by the self-healing ability of live organisms' skin surface, we report a strategy to fabricate healable underwater superoleophobic coatings. Seawater-responsive polymer-modified silica nanoparticles (PTM-SiO₂) were embedded into a self-polishing polymer (SP) matrix (Figure 1a). In the seawater, the SPs on the surface gradually

is therefore formed in situ through the development of hydrophilic components, that is, hydrolyzed SP and PTM-SiO₂. The above strategy for underwater superoleophobic coatings has two critical advantages: (1) the hydrophilic components are mainly limited to the surface, which prevents the coating from swelling with seawater; (2) the coating surface is always in a dynamic state because of the constant self-peeling of the binder resin, favoring the detachment of a variety of pollutants. In contrast to previous reports on underwater superoleophobic surfaces, this approach is quite simple and the coating is capable of mimicking the self-renewal ability of living creatures in seawater, without any additional conditions.

Currently, SPs are mainly employed to fabricate tin-free self-polishing antifouling coatings;^{26–28} however, these coatings generally contain the antifoulant cuprous oxide microparticles, which are harmful to the marine environment. In addition, they do not possess underwater superoleophobic surfaces. Our work could possibly provide a new biomimicking approach to overcome the toxicity of conventional marine antifouling coatings.

2. RESULTS AND DISCUSSION

2.1. Synthesis of Seawater-Responsive Silica Nanoparticles. Triisopropylsilyl acrylate (TISPA), a common functional monomer used in the synthesis of self-polishing resins,^{29,30} was chosen to synthesize seawater-responsive poly(TISPA-co-MPS) (PTM), together with 3-methacryloxypropyltrimethoxysilane (MPS), via radical polymerization. The PTM polymer was then grafted onto the silica nanoparticles via siloxane condensation, as schematically shown in Figure S1. Silica nanoparticles with a primary size of 20 nm were chosen based on our previous experience on fabrication of underwater superoleophobic coatings.²² The as-obtained PTM-SiO₂ nanoparticles were characterized by Fourier-transform infrared (FT-IR) spectroscopy, thermogravimetric analysis (TGA), and wetting behavior. As seen from the FT-IR spectra of Figure 2a, the characteristic peaks centered at 2949, 2866, and 1709 cm⁻¹, corresponding to the stretching vibrations of the methyl and carbonyl groups, appeared in the spectrum of PTM-SiO₂, suggesting the successful anchoring of PTM. The TGA curve of PTM-SiO₂ (Figure 2b) showed a weight loss as high as 20%, which should be caused by the thermal degradation of the grafted PTM chains. This weight loss corresponded to a thickness of approximately 1.8 nm, assuming that the density of the grafted PTM layer was 1.0 g/cm³. Both the unmodified SiO₂ and PTM-SiO₂ nanoparticles were further pressed into sheets to understand the difference in their wetting behaviors. As illustrated in Figure 2c,d, the water droplet quickly spread on the unmodified SiO₂ sheet, whereas it could not wet the PTM-SiO₂ sheet. This phenomenon also indirectly demonstrated that PTM had been grafted onto the surface of the SiO₂ nanoparticles.

The PTM-SiO₂ powder was further charged into the artificial seawater (ASW) under magnetic stirring. The particles initially kept afloat (Figure 2e(i)), then slowly sank in ASW (Figure 2e(ii)), suggesting their transition from a hydrophobic to a hydrophilic character. The PTM-SiO₂ nanoparticles that had been introduced into the ASW were centrifuged, washed with deionized water, dried overnight, and then analyzed by FTIR (Figure 2a(iv)). Compared to the PTM-SiO₂, a new peak at 1565 cm⁻¹, assigned to the stretching vibration of

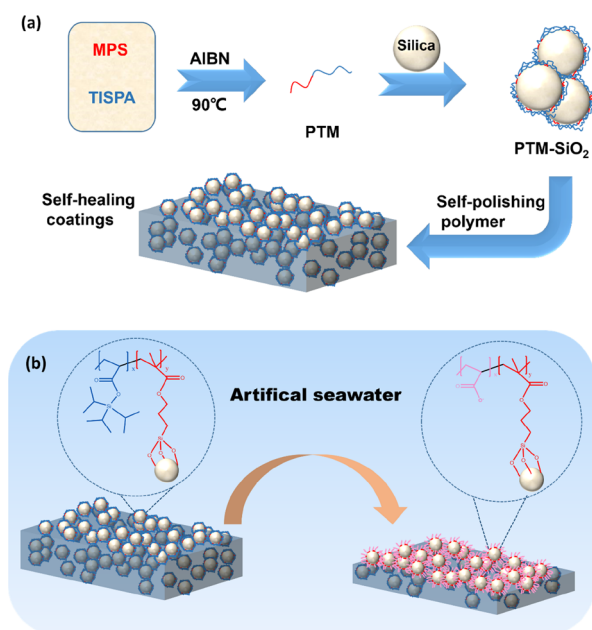


Figure 1. Schematic diagram of the fabrication strategy of seawater-induced underwater superoleophobic coatings. (a) Fabrication procedure of the coating. (b) Formation mechanism of an underwater superoleophobic surface in seawater based on SP/PTM-SiO₂ coatings.

hydrolyze, acquiring high hydrophilicity, and ultimately dissolve into the seawater (Figure 1b). The self-peeling of the SP leads to the exposure of the PTM-SiO₂ on the surface, which undergoes further seawater-triggered hydrolysis to develop a hydrophilic character through the transformation of the silyl ester bonds of the PTM into carboxylate ions groups. Because the self-peeling of the hydrolyzed PTM-SiO₂ is slower than that of SP, a relatively high fraction of nanoparticles is produced at the surface, thus establishing the micro/nanostructure. The underwater superoleophobic surface

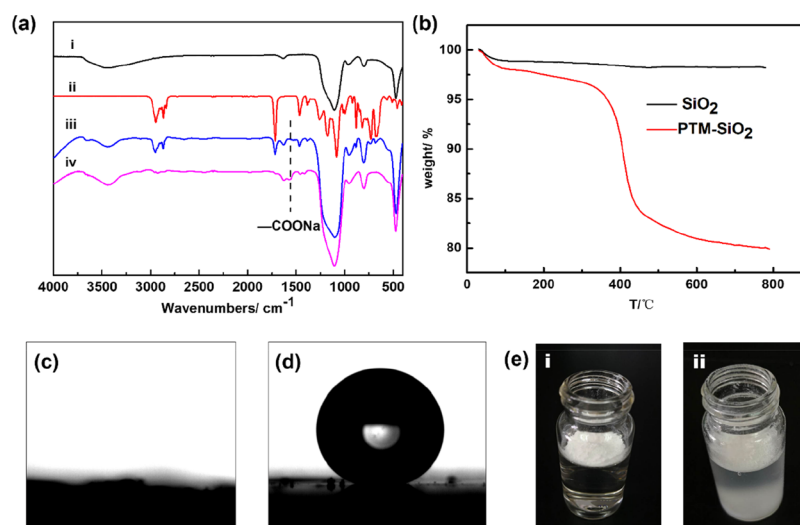


Figure 2. (a) FT-IR spectra of (i) unmodified SiO_2 , (ii) PTM, (iii) PTM- SiO_2 , and (iv) hydrolysis product of PTM- SiO_2 in ASW; (b) TGA curves of unmodified SiO_2 and PTM- SiO_2 ; water droplet on the sheet of (c) unmodified SiO_2 and (d) PTM- SiO_2 ; (e) PTM- SiO_2 powder (i) immediately after introduction into ASW and (ii) after stirring for 2 days.

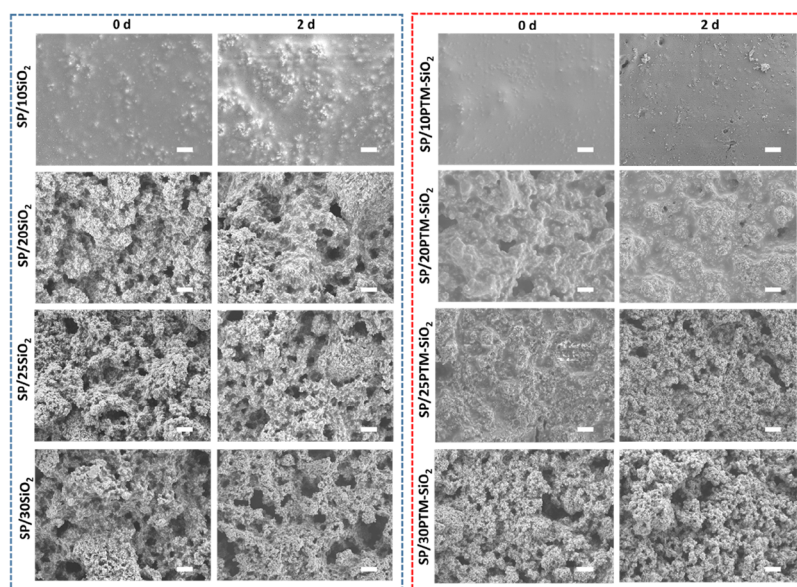


Figure 3. FE-SEM images of the surface of the SP/ $x\text{SiO}_2$ and SP/ $x\text{PTM-SiO}_2$ coatings before and after submersion in ASW for 2 days. The scale bar is 1 μm .

$-\text{COONa}$, appeared, indicating the hydrolysis of the triisopropylsilyl ester group. To further demonstrate the hydrolysis mechanism, the proton nuclear magnetic resonance (^1H NMR) spectra of PTM before and after hydrolysis were compared (Figure S2). The peak attributed to the methyl groups of the isopropylsilyl groups weakened markedly after hydrolysis, demonstrating that in ASW the silyl ester bonds were hydrolyzed to carboxylate. The results showed that the PTM- SiO_2 nanoparticles responded to the seawater environment as desired.

2.2. Morphology of Seawater-Responsive Coatings.

Unmodified SiO_2 and PTM- SiO_2 nanoparticles were used to fabricate SP/ $x\text{SiO}_2$ and SP/ $x\text{PTM-SiO}_2$ coatings (where x denotes the nanoparticle content), respectively. The coatings were sprayed on polyethylene glycol terephthalate (PET) sheets ($7.5 \times 2.5 \text{ cm}^2$) and dried at room temperature. It was found that many cracks appeared on the surface of the SP/

$x\text{SiO}_2$ coatings, but not on the surface of the SP/ $x\text{PTM-SiO}_2$ coatings (Figure S3). This suggested that the PTM- SiO_2 nanoparticles had a higher wettability with SP. The surface morphology of these coatings was further inspected by field emission scanning electron microscopy (FE-SEM), as shown in Figure 3. It is obvious that the surface roughness gradually increased with increasing nanoparticle content for both the SP/ $x\text{SiO}_2$ and SP/ $x\text{PTM-SiO}_2$ coatings. The SP/ $x\text{SiO}_2$ coatings displayed a porous surface even with a SiO_2 content as low as 20%. On the other hand, the SP/ $x\text{PTM-SiO}_2$ coatings exhibited a porous surface only at a PTM- SiO_2 content of 30%. The phenomenon indicated a higher critical pigment volume concentration of SP/PTM- SiO_2 coatings compared to the SP/ SiO_2 coatings, which confirmed the better dispersion of PTM- SiO_2 nanoparticles in the SP.^{31,32}

After submerging the coatings in ASW for 2 days, new surface morphologies were revealed for some of the coatings.

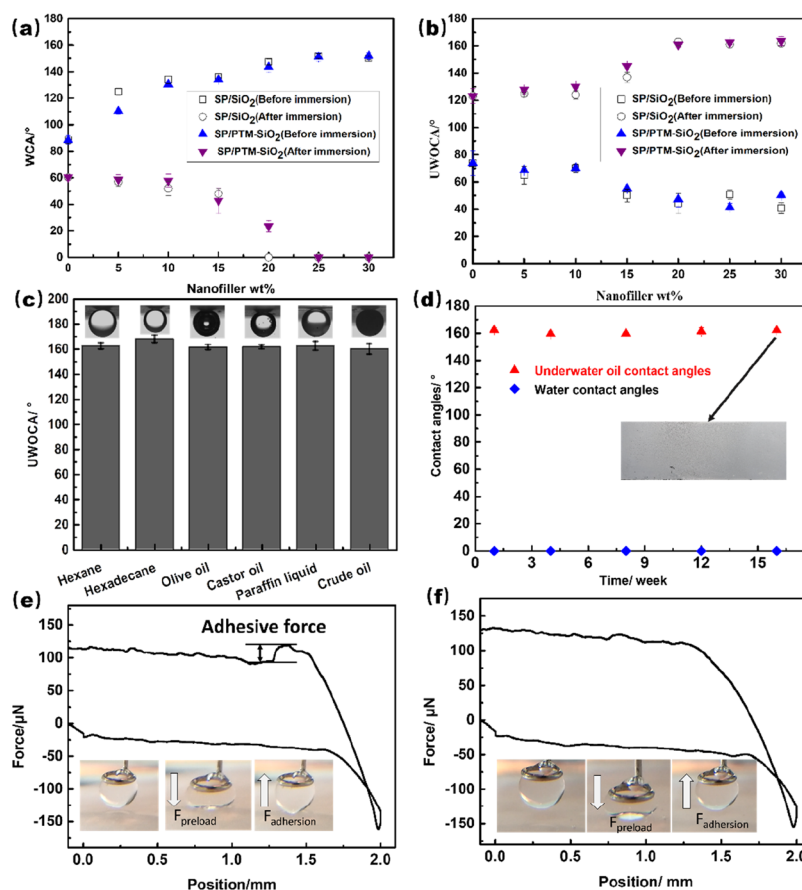


Figure 4. Wetting behavior and oil-repellence of SP/ x SiO₂ and SP/ x PTM-SiO₂ coatings. (a,b) WCA and UWOCAs of SP/ x SiO₂ and SP/ x PTM-SiO₂ coatings before and after immersion in ASW for 2 days; (c) UWOCAs of various oil droplets on the surface of SP/30PTM-SiO₂; (d) UWOCAs and WCA of SP/30PTM-SiO₂ during immersion in ASW for 16 weeks; the inset shows that the coating remained intact after 16 weeks; the adhesive force of *n*-decane was (e) $19.1 \pm 3.7 \mu\text{N}$ on the SP/30SiO₂ surface and (f) negligible on the SP/30PTM-SiO₂ surface.

Particularly, the relatively dense SP/25PTM-SiO₂ coating became porous, thus creating a distinctive micro/nanostructure. Increased surface roughness was observed for the SP/10SiO₂, SP/10PTM-SiO₂, and SP/20PTM-SiO₂ coatings. All these topographical surface changes should be attributed to the dissolution of the SP resin in ASW. On the other hand, the coatings that were originally porous did not show remarkable changes in surface morphology after submersion in ASW.

2.3. Wetting Behavior of Seawater-Responsive Coatings. The water contact angles (WCAs) in air and the underwater oil contact angles (UWOCAs) with *n*-decane probe liquid were determined, as shown in Figure 4a,b. For both the SP/ x SiO₂ and SP/ x PTM-SiO₂ coatings, the WCA increased with the nanofiller content. The coatings achieved a superhydrophobic surface (WCA > 150°) when the SiO₂ (or PTM-SiO₂) content exceeded 20%. The UWOCA slightly decreased with increasing SiO₂ (or PTM-SiO₂) content. None of the coatings acquired underwater superoleophilicity, regardless of the silica content or the type of silica nanoparticles; however, interestingly, after submersion in ASW for 2 days, the initial superhydrophobic coatings became underwater superoleophobic (UWOCAs > 150°). Furthermore, the underwater superoleophobic coatings possessed superhydrophilicity in air, with the exception of the SP/20PTM-SiO₂ coatings. A considerable increase of the UWOCAs was also seen for the other coatings. Besides *n*-decane, the SP/30PTM-SiO₂ coating also exhibited underwater superoleopho-

bic behavior toward a variety of oils (Figure 4c), such as hexane, hexadecane, castor oil, paraffin liquid, and crude oil. Moreover, its underwater superoleophobic performance and superhydrophilicity were maintained even after immersion in ASW for 16 weeks (Figure 4d).

Although SP/ x SiO₂ or SP/ x PTM-SiO₂ coatings were underwater superoleophobic at a nanoparticle content of 20%, a low oil adhesive force was qualitatively observed for *n*-decane and crude oil (Figure S4e,f) only when the nanoparticle content was increased beyond 25%. Quantitative measurements showed that the SP/30SiO₂ coating had an adhesive force of $19.1 \pm 3.7 \mu\text{N}$ (Figure 4e) toward *n*-decane, whereas the adhesive force of the SP/30PTM-SiO₂ coating was below the detection limit of the instrument (Figure 4f). This demonstrated that PTM-SiO₂ nanoparticles worked better than unmodified SiO₂ nanoparticles as nanofiller of the underwater superoleophobic coatings.

Combined with the observed evolution of surface morphology, the seawater-responsive wetting behavior of the coatings can be explained as follows: after submersion in ASW, the superficial SP was hydrolyzed and dissolved, resulting in the exposure of the unmodified SiO₂ or PTM-SiO₂ nanoparticles on the surface and, hence, an increased surface roughness. Meanwhile, the hydrolysis of SP also caused an increased hydrophilicity of the superficial SP. As a result, an underwater superoleophobic surface was formed when the amount of the exposed nanoparticles was above a critical value.^{9,19} For the

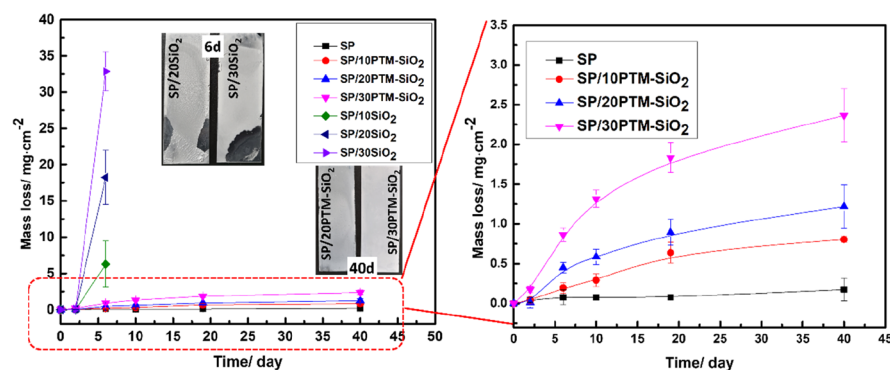


Figure 5. Mass loss of SP, SP/ x SiO₂, and SP/ x PTM-SiO₂ in ASW.

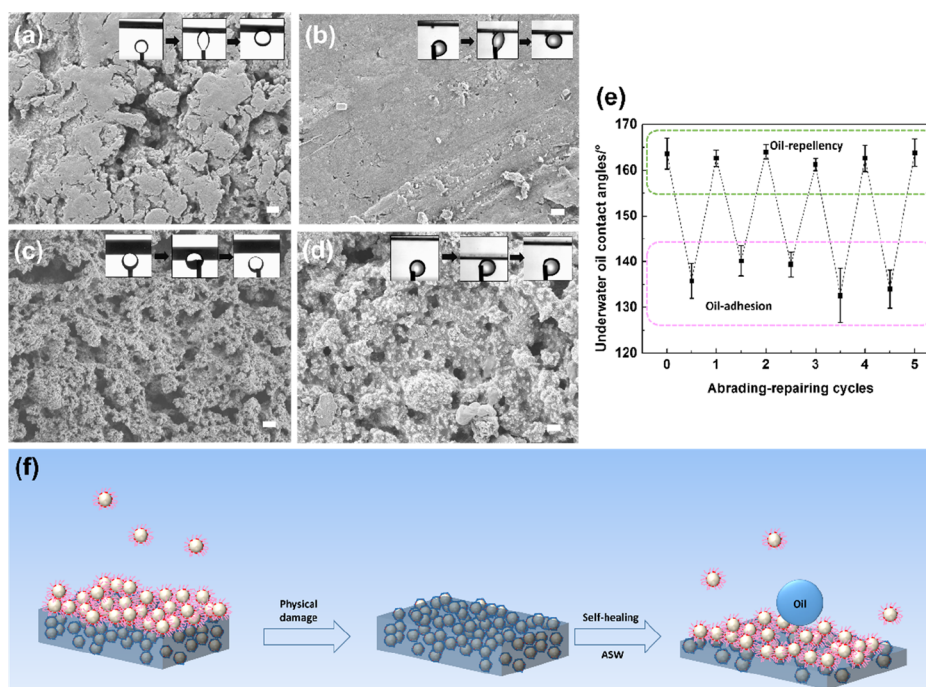


Figure 6. Seawater-induced restorability of SP/ x SiO₂ and SP/ x PTM-SiO₂ coatings. Scanning electron micrographs of the abraded surface morphology of (a) SP/30SiO₂ and (b) SP/30PTM-SiO₂; surface structure of (c) SP/30SiO₂ and (d) SP/30PTM-SiO₂ after submersion in ASW for 2 days. Insets of (a–d) show *n*-decane droplet adhesive behavior before and after recovery in ASW. Scale bar is 1 μ m. (e) Changes of UWUCA of SP/30PTM-SiO₂ during the abrading and repairing cycles. (f) Schematic diagram of the seawater-induced healing mechanism of SP/ x PTM-SiO₂ coatings.

coatings with PTM-SiO₂ nanoparticles, the grafted PTM was transformed into a hydrophilic component following its hydrolysis in ASW. The hydrolyzed PTM additionally strengthened the bonding water layer at the surface, improving the oil-repulsion behavior of the coatings.

2.4. Durability of the Coatings. The degradation rate of the coatings in ASW was determined in term of mass loss, as shown in Figure 5. It can be clearly seen that the degradation rate increased with the nanofiller content for both the SP/ x SiO₂ and SP/ x PTM-SiO₂ coatings. Moreover, the mass loss of the SP/ x SiO₂ coatings was much faster than that of the SP/ x PTM-SiO₂ coatings. The SP/20SiO₂ and SP/30SiO₂ coatings were partially removed (the inset in Figure 5) after submersion in ASW for 6 days, presumably because of the microcracks (Figure S3) and hydrophilic silica nanoparticles formed on their surfaces, both of which could accelerate their hydrolysis and decrease their adhesive force. On the contrary, the SP/20PTM-SiO₂ and SP/30PTM-SiO₂ coatings remained intact

even after 40 days' immersion (inset of Figure 5), demonstrating their superior durability. The thickness reduction of the SP/ x PTM-SiO₂ coatings in ASW was also monitored (Figure S5). Similar to the behavior of the mass loss, a higher PTM-SiO₂ content led to faster reduction in coating thickness. A thickness reduction of approximately 18 μ m (average of about 0.45 μ m/day) was observed for the SP/30PTM-SiO₂ coating after 40 days of immersion. Given the absence of a correlation between the degradation rate and the coating thickness, a 1 mm thick coating could have a service life of at least 6 years under the same conditions as those considered here. The service life could be further prolonged by lowering the degradation rate of the SP.

The stability of the SP/30PTM-SiO₂ coating at various pH values was examined. The coating was capable of maintaining its underwater superoleophobicity (UWUCA = $160.0 \pm 1.2^\circ$) for 2 h in a NaOH solution (pH = 14) and for 10 days in a H₂SO₄ solution (pH = 1) (Figure S6). The coating was also

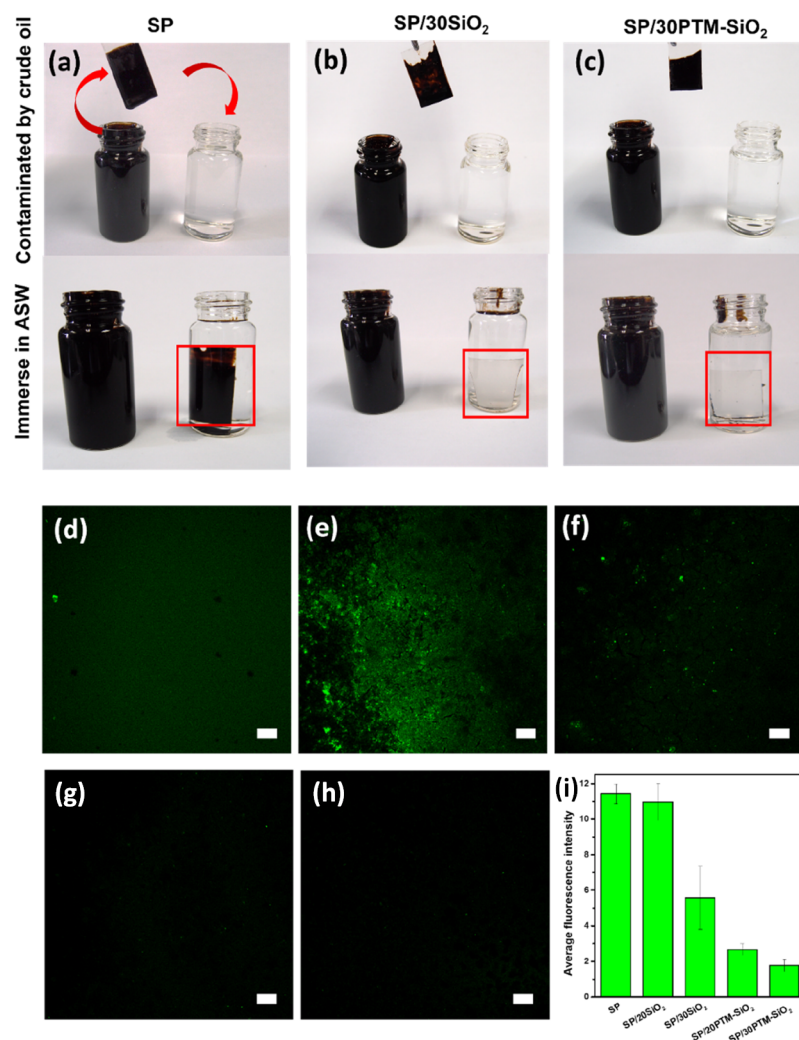


Figure 7. (a–c) Self-cleaning tests after crude oil contamination for (a) SP, (b) SP/30SiO₂, and (c) SP/30PTM-g-SiO₂ coatings. Black and transparent fluids in cuvettes are crude oil and ASW, respectively. Prewetted coatings were submerged in crude oil then placed into ASW. Confocal laser scanning microscope images of the coatings after FITC-labeled BSA adsorption tests (d) SP, (e) SP/20SiO₂, (f) SP/30SiO₂, (g) SP/20PTM-SiO₂, (h) SP/30PTM-SiO₂; the scale bar is 100 μ m. (i) Average fluorescence intensity of the FITC-BSA on coatings.

immersed into various surfactant solutions, including alkyl-phenol ethoxylates (OP-10), hexadecyl trimethyl ammonium bromide (CTMAB), and sodium dodecyl sulfate (SDS). No noticeable change was found in the UWOC value after 10 days' immersion. In addition, the coating maintained its underwater superoleophobic performance and low oil adhesive force after 10 days of UV-accelerated aging test. All these results suggested the good stability of the SP/30PTM-SiO₂ coating under various harsh environments, which is beneficial to its durability during service.

2.5. Seawater-Induced Restorability. Naturally occurring coatings show that a self-recovering ability is the best strategy to maintain their original performance. In the present work, we examined the seawater-induced restorability of the SP/30SiO₂ and SP/30PTM-SiO₂ coatings after mechanical abrasion by sandpaper. It was evident that, as a result of the mechanical abrasion, the coatings became smooth and lost their underwater superoleophobic performance (Figure 6a,b); as a consequence, *n*-decane oil droplets adhered to these damaged surfaces easily (insets of Figure 6a,b). However, after being immersed in ASW for 2 days, both the SP/30SiO₂ and SP/30PTM-SiO₂ coatings recovered their surface micro/

nanostructures and exhibited a low oil adhesive force (Figure 6c,d). Furthermore, the SP/30PTM-SiO₂ surface could repeatedly restore its underwater superoleophobicity even after 5 cycles of mechanical abrasion and immersion in ASW (Figure 6e), indicating its excellent restorability. The damaged surface of the coatings can also be recovered in diluted ammonia solution with a pH value of 9 (Figure S7). In contrast, the underwater superoleophobic surface of the abraded samples could not be recovered by submersion in deionized water or dilute sulfuric acid solution with a pH value of 5 (Figure S7). Figure 6f schematically describes the seawater-induced healing mechanism. The seawater-triggered hydrolysis takes place on the abraded coating surface once it is exposed, because of the weak alkaline character of seawater. When a critical degree of hydrolysis is reached, the SP chains acquire a sufficiently high hydrophilicity and dissolve in seawater. Although PTM-SiO₂ nanoparticles also undergo hydrolysis and acquire a hydrophilic surface, they come in contact with seawater after the complete dissolution of the SP binders surrounding them. This meant that the self-peeling of nanoparticles from the top surface layer of coatings in seawater is slower for PTM-SiO₂ than SP nanoparticles. As a result, the

volume fraction of PTM-SiO₂ nanoparticles at the surface increases, resulting in reconstruction of the surface micro/nanostructure. Furthermore, the underwater superoleophobic surface was restored as a consequence of the hydrolysis of the SP and PTM-SiO₂ nanoparticles. According to this mechanism, the SP/30PTM-SiO₂ coatings will maintain their underwater superoleophobicity until the film has completely disappeared, meaning that they will be capable of carrying out their protective function during their entire service life. This feature is favorable for their practical application.

2.6. Self-Cleaning and Antifouling Abilities. In marine environments, oil contamination and silt sediments seriously weaken the antifouling efficacy of coatings. A self-cleaning ability of the coatings is therefore required. In this work, crude oil was used as contaminant to assess the self-cleaning properties of the coatings. The coating panel was pretreated with ASW, stained with crude oil, and placed into ASW. The crude oil was easily peeled off from the panels coated with either SP/30SiO₂ (Figure 7b) or SP/30PTM-SiO₂ coatings (Figure 7c), suggesting their excellent self-cleaning properties. In contrast, crude oil could not be removed from the SP-coated panel in ASW (Figure 7a). Also, slight contamination took place on the SP/30SiO₂ and SP/30PTM-SiO₂ coatings with physically damaged surfaces (Figure 7a,b); however, if the abraded coatings were submerged in ASW for 2 days, their self-cleaning properties were completely recovered (Figure S7c,d). This phenomenon further demonstrated the seawater-induced healing abilities of the coatings.

Protein adsorption on material surface in seawater occurs within seconds to minutes and is regarded as the main phenomenon responsible for the formation of a conditional film allowing the settlement of micro-organisms, such as diatoms and bacteria,³³ as well as secondary macro-organisms,³⁴ such as algal spores^{35–37} and larvae of barnacles and tubeworms.^{38,39} Therefore, contrasting protein adsorption could be one of the ways to achieve an antifouling behavior. We employed fluorescein isothiocyanate-labelled bovine serum albumin (FITC-BSA) as a model foulant to evaluate the antiprotein adsorption characteristics of the coating (see details in the experimental section). Figure 7d–h shows the confocal laser scanning microscope images of different coatings after FITC-BSA adsorption tests. Their average fluorescence intensities were also measured and are shown in Figure 7i. For the SP coating, brilliant fluorescence was observed, indicating that a high amount of FITC-BSA had been adsorbed on the surface. Surprisingly, the amount of FITC-BSA adsorbed on the SP/20SiO₂ coatings was similar to the amount adsorbed on the SP coating. This suggested that the nano-SiO₂-based micro/nanostructure promotes protein adsorption. The protein adsorption on the SP/30SiO₂ coating decreased compared to the SP/20SiO₂ coating, presumably resulting from the slight surface chalking of the SP/30SiO₂ coating. The chalked surface layer easily fell off during the protein adsorption tests. In contrast to the strong protein adsorption observed on the SP/*x*SiO₂ coatings, the SP/*x*PTM-SiO₂ fluorescence intensities noticeably decreased, indicating a significantly higher resistance to protein adsorption. This result was attributed to the hydrolyzation of PTM-SiO₂ on surface, which captured high amounts of bonding water, thus contributing to the resistance to protein adhesion.

Previous studies have shown that the resistance to protein adsorption is determined by the hydration and steric repulsion.^{40–43} The significantly higher antiprotein perform-

ance of the SP/*x*PTM-SiO₂ coatings compared to the SP/*x*SiO₂ coatings was thus explained as follows: the silyl ester group in PTM is converted to the carboxylate group, providing a hydrogen bond acceptor; the hydrophilic segments hydrate in seawater, forming a layer of water molecules bound to the material's surface via hydrogen bonds. These bonds were much stronger than the water–water bonds. For protein adsorption to take place on the SP/*x*PTM-SiO₂ coating, the proteins need to simultaneously replace the interfacial water and overcome the conformational entropy hinderance offered by the hydrophilic polymer chains, which is a very difficult process. On the other hand, only a few hydrophilic polymer chains are present on the surface of the SP/SiO₂ coating, thus limiting the entropy hinderance at molecular level for protein adsorption. In addition, the silanol groups dominating on the SP/SiO₂ coating surface are hydrogen bond donors, whereas the carboxylate groups on the PTM-SiO₂ surface are hydrogen acceptors. Generally, the presence of hydrogen acceptors and the absence of hydrogen bond donors contributed to the resistance to protein adsorption.^{43–45} Hence, the SP/*x*PTM-SiO₂ coatings performed better in antiprotein adsorption.

3. CONCLUSIONS

In this study, a seawater-induced strategy to fabricate healable underwater superoleophobic antifouling coatings was developed. Nanocomposite coatings were prepared by blending SP with either unmodified or PTM-grafted SiO₂ nanoparticles and submerged in ASW. Regardless of the type of SiO₂ nanoparticles, underwater superoleophobic surfaces were attained for SiO₂ nanoparticle contents higher than 20%. The coatings with PTM-SiO₂ nanoparticles had a longer service life, much lower oil adhesive force, and higher resistance to protein adsorption, compared to those with unmodified SiO₂ nanoparticles. These differences were due to the presence of hydrolyzed PTM on the surface of the coatings, which caused the formation of a layer of water molecules strongly bound to the material surface. This seawater-induced strategy allowed the simultaneous in situ production of a surface micro/nanostructure and of hydrophilic chains through the hydrolysis of superficial SP and of the grafted PTM polymer. This mechanism of formation of an underwater superoleophobic surface endowed the coatings with a seawater-triggered healing capability that well mimicked live fish scales, while also preventing the severe water swelling that generally takes place on underwater superoleophobic coatings based on hydrophilic polymers. These advantages of the seawater-induced strategy could assure the long-term durability of the coatings in seawater. By simply adjusting the structure of the grafted polymer on the SiO₂ nanoparticles or the hydrolysis rate of SP, these coatings are expected to meet the antifouling requirements when applied to microfluids, oil/water separation, marine equipment, and ships.

4. EXPERIMENTAL SECTION

4.1. Materials. The SP with a commercial name of H100 (a solution of silicon-based self-polishing resin in xylene, *M_w* = 16 000 g/mol, polydispersity index = 1.4, solid content 55%) was provided by Shenzhen Haiwei New Materials Technology Co. Ltd. (Shenzhen, China). Silica nanoparticles (20 nm) were purchased from Nanjing Haitai Nano Materials Co. Ltd. (Nanjing, China). Xylene and MPS were purchased from Aladdin Chemical Reagent (China). TISPA was purchased from Hubei Jusheng Technology Co., Ltd. (China). Deuterated chloroform (CDCl₃, D, 99.8% + 0.03% v/v tetramethylsi-

lane) was purchased from Shanghai Macklin Biochemical Co., Ltd. (China). Acetone- d_6 (D, 99.9%) was purchased from Cambridge Isotope Laboratories, Inc. ASW (pH: 8.2) was prepared according to ASTM D1141-90. The solution contains NaCl (24.53 g/L), $MgCl_2$ (5.2 g/L), Na_2SO_4 (4.09 g/L), $CaCl_2$ (1.16 g/L), KCl (0.695 g/L), $NaHCO_3$ (0.201 g/L), KBr (0.101 g/L), H_3BO_3 (0.027 g/L), $SrCl_2$ (0.025 g/L), and NaF (0.003 g/L). FITC-BSA was purchased from Beijing Bersee Science and Technology Co. Ltd (Beijing, China).

4.2. Preparation of PTM. PTM was prepared by radical polymerization. Xylene (50 mL) was added to a four-necked flask, and nitrogen was passed through the solvent for 40 min to remove oxygen. The solvent was heated to 90 °C and a mixture of deoxygenated TISPA (50 g), MPS (10 g), xylene (60 g), and AIBN (0.8 g) was then dropped into the four-necked flask under mechanical stirring for 4 h. After dropping, the mixture was stirred at a fixed temperature for an additional 4 h. Thereafter, the reactant was purified with methanol three times to obtain the formed PTM. The 1H NMR spectrum of the PTM showed that the molar ratio of MPS to TISPA was 1:10 (Figure S2a). Gel permeation chromatography tests showed that its number average molecular weight was 3366 g/mol (Figure S9).

4.3. Fabrication of PTM-SiO₂. SiO₂ nanoparticles (10 g) were dispersed in 100 mL of xylene under mechanical stirring at 50 °C, followed by ultrasonication for 10 min. Then, the dispersion, together with 5 g of PTM, was charged into a 250 mL round-bottomed flask. The mixture was further stirred at 50 °C for 12 h. After that, the SiO₂ nanoparticles were extracted by centrifugation at 10 000 rpm and then washed with xylene six times to remove the free PTM. The particles were then placed in a vacuum oven at 50 °C for 2 days, resulting in the formation of PTM-SiO₂. In order to measure the surface wettability of the nanoparticles, sheets of both unmodified SiO₂ and PTM-SiO₂ nanoparticles were fabricated by placing the nanopowder (about 0.1 g) between two glass slides and loading the glass slides with a 500 g counter poise for 20 min.

4.4. Preparation of SP/ x PTM-SiO₂ (or SiO₂) Coatings. PTM-SiO₂ and SP were blended in xylene under magnetic stirring for 4 h, and followed by ultrasonic treatment for 30 min. The solid content of the coatings was fixed at 40 wt %. Then, the blend was sprayed on a PET sheet (7.5 × 2.5 cm²) using a SATA spray gun (model: minijet 4400-120) at a spray distance of ~20 cm and spray pressure of 0.5–0.6 MPa, and dried at room temperature to get SP/ x PTM-SiO₂ coatings. SP/ x SiO₂ coatings were prepared by the same method using unmodified SiO₂ nanoparticles as the nanofiller. The x in the sample names of SP/ x PTM-SiO₂ and SP/ x SiO₂ represents the weight percentage of unmodified SiO₂ or PTM-SiO₂ in the dry coating. The thickness of all coatings was 45 ± 5 μm, measured by an optical microscope.

4.5. Characterization. FTIR analysis was carried out with a Bruker VERTEX 70 spectrometer. Prior to characterization, the sample was fabricated as a KBr disc. TGA measurements were carried out on a Q500 instrument (TA Instruments) with an air flux of 40 mL/min and at a heating rate of 10 °C/min. The morphology of the coatings was observed by FE-SEM (Zeiss, Ultra 55, Oberkochen, Germany) at an accelerating voltage of 3 kV. The WCA and UWOC were measured using a contact angle analyzer (OCA25, Dataphysics, Germany). The average value from at least five parallel measurements with a 3 μL liquid droplet was considered as the final result. Molecular weight and its distribution were determined using a gel permeation chromatograph (Shimadzu, LC-10ADvp, Japan). Tetrahydrofuran was used as the eluting solvent and testing parameter, at a flow rate of 0.8 mL/min and a temperature of 35 °C, and monodispersed polystyrene was used as the calibration standard. The 1H NMR spectrum was recorded on a Bruker (400 MHz) NMR spectrometer, using $CDCl_3$ and acetone- d_6 as solvents for PTM and hydrolyzed PTM, respectively. The hydrolyzed PTM was obtained as follows: 30 mg of PTM, 1 mL of acetone, and 40 μL of ASW were mixed and stored at room temperature for 2 days. The undissolved solid was then removed by centrifugation, and the residual solution was evaporated to get the hydrolyzed PTM.

4.6. Degradation Rate Test. The coatings were sprayed on PET substrates (size: 7.5 × 2.5 cm²) and dried at room temperature for 1

day. These samples were then submerged in ASW. At fixed time intervals, the samples were taken out and soaked in deionized water for 10 min. The coatings were then rinsed with deionized water and dried in an oven at 50 °C for 6 h. The weight of the dry coatings was recorded. The mass loss was calculated as $(W_0 - W_t)/A$, where W_0 and W_t are the initial sample weight and the weight of the dry coating after immersion in ASW, respectively, and A is the coating area.

4.7. Stability Tests under Harsh Environments. The SP/30PTM-SiO₂ coating on the PET substrate was immersed in aqueous solutions of H_2SO_4 (0.5 mol/L, pH = 1), NaOH (1.0 mol/L, pH = 14), SDS (0.25 g/mL), CTMAB (0.25 g/mL), and alkylphenol ethoxylates (OP-10, 0.25 g/mL) at ambient temperature. The coating was also exposed to ultraviolet radiation in a QUV-accelerated weathering tester (QUV/se, Q-PANEL Co. Ltd., USA). The testing cycle was set as follows: (i) UV irradiation stage: UV irradiation intensity of 0.71 W/m² (wavelength, 310 nm), testing chamber temperature of 60 °C, and duration time of 4 h, and (ii) condensation stage: testing chamber temperature of 50 °C and duration time of 4 h. At fixed time intervals, the sample was taken out and rinsed with deionized water. Thereafter, the UWOC values were recorded.

4.8. Seawater-Induced Healing Test. A coated PET sheet (size: 7.5 × 2.5 cm²) was placed on sandpaper (1500 mesh), with the coated side in contact with the sandpaper. A 500 g counter poise with a diameter of 27 mm was then loaded on the PET sheet. The load moved back and forth within a distance of 10 cm four times at a speed of about 1 cm/s. After that, the sample was submerged in ASW for 2 days. The UWOC values of the abraded coating were recorded before and after immersion in ASW, and the surface morphology was observed with FE-SEM. For the sake of comparison, the abraded coatings were also immersed in deionized water, diluted sulfate solution (pH = 5), and diluted ammonia solution (pH = 9), respectively. Similarly, their surface wetting behavior underwater as well as their surface morphology was checked after immersion for 2 days.

4.9. Protein Adsorption Test. The coating samples with a size of 2.5 × 2.5 cm² were submerged in the FITC-BSA solution (20 μg/mL) at 25 °C for 8 h in the dark. The panels were rinsed with PBS (20 mL) to remove the nonadhesive protein, then dried at room temperature in the dark. The absorption of the FITC-BSA on coatings was observed by a confocal laser scanning microscope (Nikon C21, Tokyo, Japan) using a 488 nm laser excitation, and the average fluorescence intensity was calculated by the ImageJ software.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.8b16464.

Synthetic route of PTM-SiO₂; 1H -NMR spectra of PTM and hydrolyzed PTM; optical images of SP/ x SiO₂ and SP/ x PTM-SiO₂ coatings; optical images of *n*-decane droplets on SP/30SiO₂ and SP/30PTM-SiO₂ surface; reduction of coating thickness in ASW; time dependence of underwater oil contact angles of SP/30PTM-SiO₂ coating immersed in various solutions; the healing performance of SP/30PTM-SiO₂ in deionized water, dilute sulfuric acid and ammonia solution; crude oil removal performance of abraded SP/30SiO₂ and SP/30PTM-SiO₂; and GPC curve of PTM (PDF)

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Notes

The authors declare no competing financial interest.

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